

KAHLER ASTEN, Germany

WMO: 104270

Lat: 51.1803N

Lon: 8.4892E

Elev: 859

StdP: 91.43

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-14.7	-11.6	-19.7	0.7	-10.9	-16.5	1.0	-10.4	15.9	3.7	14.3	-0.7	6.5	90	0.804

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.5	23.7	16.3	22.1	15.3	20.5	14.6	17.1	21.9	16.2	20.6	15.4	19.2	4.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.4	12.1	18.6	14.5	11.5	17.6	13.7	10.9	16.8	51.4	21.7	48.6	20.8	46.0	19.3	20.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.4	12.1	11.0	-14.8	27.0	4.7	1.6	-18.2	28.2	-21.0	29.1	-23.6	30.0	-27.0	31.2
			-15.3	19.2	4.6	0.9	-18.6	19.8	-21.3	20.3	-23.9	20.8	-27.3	21.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.5	-2.3	-2.7	0.7	3.6	8.8	11.4	13.8	13.7	10.4	6.6	1.7	-0.4		
	DBStd	7.09	4.66	4.96	4.05	4.10	4.26	4.10	3.76	3.58	3.45	3.62	4.03	3.46		
	HDD10.0	2069	382	357	290	194	76	32	7	7	36	118	250	322		
	HDD18.3	4705	640	590	548	441	295	211	148	149	239	364	500	580		
	CDD10.0	422	0	0	1	3	39	73	124	122	47	12	1	0		
	CDD18.3	15	0	0	0	0	0	2	7	6	0	0	0	0		
	CDH23.3	63	0	0	0	0	0	6	29	28	0	0	0	0		
	CDH26.7	4	0	0	0	0	0	0	1	4	0	0	0	0		
(14) Wind	WSAvg	6.2	7.5	7.1	6.6	5.8	5.3	5.1	5.2	5.1	5.7	6.5	6.9	7.1		
(15) Precipitation	PrecAvg	1105	115	91	84	64	78	84	94	91	87	90	106	120		
	PrecMax	1492	255	222	191	118	176	138	160	202	212	308	189	287		
	PrecMin	854	4	24	13	0	33	43	40	21	23	24	3	27		
	PrecStd	171	60	52	48	31	35	25	34	46	45	53	46	58		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.3	9.9	13.9	16.8	22.2	24.1	25.8	26.6	22.3	18.2	12.7	8.7		
		MCWB	3.3	5.4	7.6	9.7	13.5	16.6	16.9	17.0	14.5	13.0	7.4	6.4		
	2%	DB	5.7	7.4	10.5	13.9	19.7	21.9	24.0	23.3	19.8	15.0	10.5	7.0		
		MCWB	3.5	4.9	6.0	8.1	11.9	15.3	16.7	16.1	14.2	11.3	6.9	5.7		
	5%	DB	4.4	5.3	7.8	12.0	17.7	20.0	22.1	21.5	17.4	13.2	8.5	5.7		
		MCWB	3.1	3.5	5.3	7.0	11.2	14.3	15.7	15.4	13.3	10.6	6.8	4.5		
	10%	DB	3.1	3.4	5.9	9.8	15.8	18.0	20.2	19.6	15.3	11.4	7.2	4.4		
		MCWB	2.2	2.3	4.3	6.0	10.4	13.2	14.9	14.5	12.4	9.9	5.9	3.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.1	7.2	8.2	10.3	14.1	17.9	18.4	18.2	16.6	13.9	9.4	7.5		
		MCDB	6.2	8.7	11.4	15.0	19.1	22.1	23.4	24.7	19.7	16.2	10.4	7.8		
	2%	WB	4.7	5.4	6.8	8.9	13.0	16.2	17.2	16.7	15.0	12.5	8.1	6.1		
		MCDB	4.8	6.8	9.0	12.5	17.7	20.4	22.2	21.5	18.0	14.1	9.0	6.6		
	5%	WB	3.4	3.7	5.7	7.7	12.0	15.0	16.3	15.9	13.9	11.1	7.2	5.0		
		MCDB	3.8	4.9	7.6	10.9	16.9	18.8	20.9	20.2	16.0	12.3	8.2	5.3		
	10%	WB	2.2	2.4	4.5	6.5	10.9	13.8	15.4	15.1	12.9	10.0	6.0	3.9		
		MCDB	2.7	3.3	5.9	9.3	15.1	17.4	19.4	18.5	14.9	11.1	6.9	4.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.9	4.6	4.8	6.4	7.5	6.9	7.5	7.2	6.0	4.9	3.9	3.6		
		MCDBR	5.6	7.0	7.4	9.0	10.3	9.5	10.0	9.7	8.7	7.2	6.2	5.2		
	5% WB	MCWBR	4.8	4.9	4.5	4.8	5.3	5.0	5.0	4.8	4.8	4.4	4.7	4.3		
		MCWBR	4.7	6.3	6.5	8.4	9.2	8.7	9.3	8.7	7.5	5.7	5.5	4.2		
(40) Clear-Sky Solar Irradiance	taub		0.282	0.304	0.333	0.362	0.366	0.367	0.374	0.370	0.343	0.323	0.303	0.281		
		taud	2.456	2.422	2.379	2.339	2.347	2.355	2.326	2.363	2.436	2.487	2.474	2.448		
	Ebn at Noon		759	832	866	877	888	888	873	859	847	799	721	697		
		Edh at Noon	57	77	97	113	117	117	119	109	91	72	56	50		
(44) All-Sky Solar Radiation	RadAvg		0.71	1.39	2.50	4.00	4.79	5.24	4.97	4.26	3.06	1.79	0.81	0.53		
	RadStd		0.10	0.22	0.34	0.50	0.52	0.48	0.65	0.40	0.42	0.19	0.12	0.06		

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47)	Regional (57 neighbors)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page